

AIM:

To implement the simple substitution technique named Caesar cipher using C language.

ALGORITHM:

STEP-1: Read the plain text from the user.

STEP-2: Read the key value from the user.

STEP-3: If the key is positive then encrypt the text by adding the key with each character in the plain text.

STEP-4: Else subtract the key from the plain text.

STEP-5: Display the cipher text obtained above.

PROGRAM:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <ctype.h>
```

```
int main()
```

```
{
```

```
    char plain[100], cipher[100];
```

```
    int key, i, length;
```

```
    printf("Enter the plain text: ");
```

```
    scanf("%s", plain);
```

```
    printf("Enter the key value: ");
```

```
    scanf("%d", &key);
```

```
    key = key % 26;
```

```
length = strlen(plain);
```

```
printf("\nPLAIN TEXT    : %s", plain);
```

```
printf("\nENCRYPTED TEXT : ");
```

```
for(i = 0; i < length; i++)
```

```
{
```

```
    if(isupper(plain[i]))
```

```
    {
```

```
        cipher[i] = ((plain[i] - 'A' + key) % 26) + 'A';
```

```
    }
```

```
    else if(islower(plain[i]))
```

```
    {
```

```
        cipher[i] = ((plain[i] - 'a' + key) % 26) + 'a';
```

```
    }
```

```
    else
```

```
    {
```

```
        cipher[i] = plain[i];
```

```
    }
```

```
    printf("%c", cipher[i]);
```

```
}
```

```
cipher[length] = '\0';
```

```
printf("\nDECRYPTED TEXT : ");
```

```
for(i = 0; i < length; i++)
```

```
{
    if(isupper(cipher[i]))
    {
        plain[i] = ((cipher[i] - 'A' - key + 26) % 26) + 'A';
    }
    else if(islower(cipher[i]))
    {
        plain[i] = ((cipher[i] - 'a' - key + 26) % 26) + 'a';
    }
    else
    {
        plain[i] = cipher[i];
    }

    printf("%c", plain[i]);
}

plain[length] = '\0';

return 0;}
```

OUTPUT:

```
Enter the plain text: KESHAVARAM
Enter the key value: 6

PLAIN TEXT      : KESHAVARAM
ENCRYPTED TEXT   : QKYNGBGXGS
DECRYPTED TEXT   : KESHAVARAM

...Program finished with exit code 0
Press ENTER to exit console.[]
```

RESULT:

Thus the implementation of Caesar cipher had been executed successfully.